·指南与共识·

重复经颅磁刺激治疗专家共识

中国医师协会神经调控专业委员会电休克与神经刺激学组

[摘要] 重复经颅磁刺激(repetitive transcranial magnetic stimulation, rTMS)治疗专家共识由中国医师协会神经调控专业委员会电休克与神经刺激学组专家团队编写。专家共识根据截至 2017 年 11 月发表的临床研究并基于循证医学标准,发布 rTMS 在临床中的治疗方案推荐,包括:抑郁症、疼痛、运动障碍、中风、癫痫、耳鸣、焦虑障碍、强迫症、精神分裂症、物质成瘾和睡眠障碍等疾病。专家共识介绍了临床上 rTMS 治疗的安全性和有效性,并规范了治疗流程和操作技术。旨在为从事 rTMS 治疗的临床工作人员制定培训和应用的专业标准,最大程度上提高治疗效果,促进该治疗技术未来的发展。

[关键词] 重复经颅磁刺激;导航;专家共识

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Chinese experts consensus on repetitive transcranial tagnetic stimulation

Chinese Society of ECT & Neurostimulation

[Abstract] The Chinese Neuromodulation Society experts were commissioned to written a consensus and treatment recommendation for the therapeutic use of repetitive transcranial magnetic stimulation (rTMS) from evidence published up until November 2017, regarding depression, pain, movement disorders, stroke, epilepsy, tinnitus, anxiety disorders, obsessive-compulsive disorder, schizophrenia, craving/addiction and sleep disorders, each summary clinical recommendation is presented with the substantiating peer-reviewed, published evidence supporting that recommendation. It summarizes current evidence for the safety and efficacy of the use of rTMS therapy, aims to determine how to optimize rTMS protocols and techniques to give them relevance in routine clinical practice. In addition, professionals carrying out rTMS protocols should undergo rigorous training to ensure the quality of the technical realization, guarantee the proper care of patients, and maximize the chances of success.

[Key words] Repetitive transcranial magnetic stimulation (rTMS); Navigation; Expert consensus

经颅磁刺激(transcranial magnetic stimulation, TMS)由 Barker 等[1]于 1985 年创立。基于电磁感应与电磁转换原理,用刺激线圈瞬变电流产生的磁场穿透颅骨,产生感应电流刺激神经元引发一系列生理、生化反应。作为非侵入性刺激技术,TMS 作用于人脑引起神经活动的改变,可检测到运动诱发电位、脑电活动变化、脑血流、代谢和大脑功能状态改变。其微观作用包括细胞膜电位、动作电位、神经递质、受体、突触、神经可塑性发生变化。

医生应掌握 TMS 工作原理和作用机制,并依据临床表现、实验室和影像学检查结果,设计个体化治疗方案。先根据大脑皮质功能解剖体表投影决定刺激部位,再决定刺激模式、强度、频率、间歇、疗程等。TMS 适应证已经扩大到抑郁症、癫痫、神经性头痛等[2]。

1 TMS 模式分类[2]

1.1 单脉冲刺激　刺激皮层拇指运动区,用于测定 MEP,测定治疗能量或运动皮层功能障碍定量评估。

1.2 成对脉冲刺激　同一个线圈在数十毫秒内先后发放 2 个脉冲,刺激同一脑区或 2 个不同线圈刺激不同脑区。常用于皮层兴奋性评估。

1.3 重复脉冲刺激　按照固定频率连续发放多个脉冲的刺激模式。通常用于临床治疗和暂时性兴奋

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或抑制特定皮层功能区域。具体频率参数设置依治疗或研究目的而定。

1.4　爆发模式脉冲刺激　将一种固定频率脉冲嵌套在另一种固定频率脉冲中的刺激模式。常用爆发模式有爆发模式脉冲刺激(theta burst stimulation, TBS),是将 3 个连续 50 Hz 脉冲嵌入 5 Hz 脉冲中。TBS 序列分为 2 种:连续爆发模式脉冲刺激(continuous theta burst stimulation, cTBS)抑制皮层功能,间断爆发模式脉冲刺激(intermittent theta burst stimulation, iTBS)(刺激 2 s,间隔 8 s)兴奋皮层功能。

2　TMS 治疗风险及评估[2]

癫痫及惊厥;头皮刺痛、灼热感;头颅或体腔内存在金属磁性物质(电子耳蜗、部分心脏起搏器等植入性医疗产品);听力损害;诱发癫痫或惊厥发作风险的药物。

说明:高频刺激有诱发癫痫或抽搐发作风险。但低频刺激可以用于抗癫痫治疗[3]。12 岁以下患者佩戴耳塞可以最大程度上避免噪音对听力的损害。

3　重复经颅磁刺激安全操作及治疗规范

(1)重复经颅磁刺激(repetitive transcranial magnetic stimulation, rTMS)治疗仪器属于大型用电设备,保证电压稳定、电流不会过载,避免安全隐患。

(2)签署知情同意书并完成安全筛查评估。

(3)首次治疗测定患者皮层静息运动阈值(resting motor threshold, RMT)。患者取坐位或仰卧位,使用单脉冲模式刺激利手侧(中国人多为右利手)拇指运动区皮层(M1),刺激 10 次,其中 5 次可以诱发拇指外展肌运动(诱发拇指外展肌诱发电位达到 50 微幅以上),该刺激强度能量为 RMT。RMT 存在个体差异,且治疗能量大小是根据 RMT 进行制定,是开展 rTMS 治疗的必要环节[4]。

(4)根据治疗目的选定 rTMS 强度、频率和数目。应严格限制在安全序列范围内,避免诱发癫痫风险的安全序列[4]。

(5)rTMS 治疗靶点定位。

常用治疗靶点脑区定位方法有 3 种:①先测定 M1 区,可以观测到外显运动反应进行确定,之后以 M1 区作为参照点,沿头皮各个方向进行定位;②参照国际标准脑电电极 10~20 导联系统定位(图 1);③借助脑影像导航技术定位。脑影像包括:全脑 T1、T2 结构像、各类功能像等(如脑血流、静息态、功能区激活像等)。

(6)治疗过程中不良反应的评定。

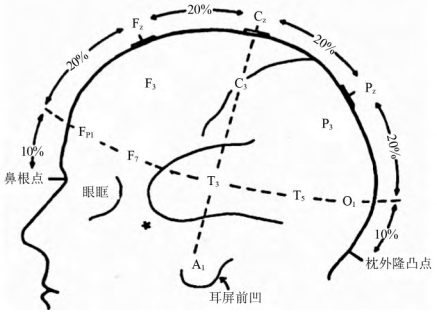

图 1　国际标准脑电电极 10~20 导联系统定位

4　rTMS 临床治疗推荐

以下内容为共识循证依据的参照标准:由欧洲神经学会联盟提出[5]。根据证据价值由高到低,分为Ⅰ~Ⅳ 4 个级别。此外还参考了牛津大学循证医学中心的循证依据标准,即优先考虑系统综述中获得的证据,也列入级别Ⅰ中。文献检索截止到 2017 年 11 月 PubMed 数据库。

4.1　抑郁症(美国食品和药品监督管理局适应证)

加拿大、新西兰、以色列、美国都已批准 rTMS 可以用于治疗抑郁症。rTMS 可以单独或联合药物治疗。但是对于病情严重,伴有自杀观念的抑郁症患者不建议单独使用 rTMS。

临床推荐 1:rTMS 高频刺激左背外侧前额叶(left dorsolateral prefrontal cortex, l-DLPFC)或低频刺激右背外侧前额叶(right dorsolateral prefrontal cortex, r-DLPFC),用于抑郁症急性期疗效肯定,连续 4~6 周,必要时可延长治疗时间。Ⅰ级一致证据[6-8]。

临床推荐 2:先前急性期 rTMS 治疗受益,目前复发的患者。Ⅰ级一致证据[6-8]。

临床推荐 3:rTMS 可作为急性期治疗获益患者的后续或维持治疗。Ⅰ级一致证据[9-10]。

临床推荐 4:rTMS 可以单独或联合抗抑郁药或其他精神类药物。Ⅱ级一致证据[8]。

4.2　慢性神经性或非神经性疼痛

临床推荐 1:rTMS 高频刺激疼痛区域对侧皮层运动区(M1)用于治疗慢性神经痛。Ⅰ级一致证据[11-17]。

临床推荐 2:rTMS 低频刺激枕叶用于治疗偏头痛(美国食品和药品监督管理局批准)。Ⅰ级一致证据。

临床推荐 3:rTMS 高频刺激 l-DLPFC 或运动皮层用于治疗非神经性疼痛,如纤维肌痛、复杂区域疼痛综合征Ⅰ型。Ⅱ、Ⅲ、Ⅳ级证据[18-19]。

4.3　运动障碍

临床推荐 1:rTMS 高频或低频刺激辅助运动皮

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层或运动皮层改善帕金森病运动症状。Ⅰ、Ⅱ、Ⅲ级证据[20-22]。

临床推荐2: rTMS高频刺激或低频刺激运动区M1或辅助运动区,用于治疗药物诱发震颤。Ⅲ级一致证据[23-24]。

临床推荐3: rTMS高频刺激l-DLPFC治疗帕金森病合并抑郁症。Ⅱ、Ⅲ级证据[25-30]。

临床推荐4: rTMS低频刺激运动区治疗肌张力障碍。Ⅲ级一致证据[20-21]。

4.4 中风

临床推荐1: rTMS高频刺激受累侧皮层运动区或低频刺激健侧皮层运动区,用于治疗运动区中风。Ⅱ、Ⅲ级证据[31-33]。

临床推荐2: rTMS高频或低频刺激布洛卡区,治疗运动性失语症。Ⅲ、Ⅳ级证据[34-35]。

临床推荐3: 爆发模式cTBS序列刺激左侧后顶叶皮层治疗偏侧忽略。Ⅲ级一致证据[36-37]。

4.5 癫痫

临床推荐1: rTMS低频刺激皮层癫痫灶治疗癫痫发作。Ⅱ、Ⅲ级证据[38-40]。

4.6 耳鸣

临床推荐1: rTMS低频刺激颞叶或颞顶叶皮层,高频刺激r-DLPFC治疗耳鸣。Ⅱ、Ⅲ级证据[41]。

4.7 焦虑障碍

临床推荐1: rTMS高频刺激r-DLPFC或低频刺激l-DLPFC治疗创伤后应激障碍。Ⅲ级一致证据[42-44]。

临床推荐2: rTMS低频刺激r-DLPFC和颞顶区治疗惊恐发作和广泛性焦虑。Ⅲ级一致证据[45-46]。

4.8 强迫症

临床推荐1: rTMS高频或低频刺激双侧DLPFC治疗强迫症。Ⅱ、Ⅲ级证据[47-50]。

4.9 精神分裂症

临床推荐1: rTMS低频刺激颞顶叶皮层治疗幻听。Ⅱ、Ⅲ级证据[51-57]。

临床推荐2: rTMS高频刺激l-DLPFC或双侧DLPFC改善精神分裂症阴性症状。Ⅱ、Ⅲ级证据[58-61]。

4.10 物质成瘾

临床推荐1: rTMS高频刺激l-DLPFC降低毒品渴求(心瘾),目前证据提示没有长期效果。Ⅱ、Ⅲ级证据[62-66]。

4.11 睡眠障碍

临床推荐1: rTMS低频1Hz刺激双侧DLPFC和顶枕区域治疗睡眠障碍。Ⅱ、Ⅲ级证据[67-68]。

5 关于临床治疗特殊群体的几点说明

5.1 孕妇 磁场强度随距离迅速衰减,所以胎儿不太可能受到TMS影响。抑郁孕妇成功接受TMS治疗而胎儿未出现副作用[69-71]。临床上孕妇接受TMS治疗应避免直接刺激腰椎。作为TMS操作者的孕妇应至少远离线圈0.7 m[72]。

5.2 儿童 研究证实单脉冲和成对脉冲TMS用于儿童安全[73]。发育过程中的因素确实影响儿童接受TMS安全性: ①皮质兴奋性水平(新生儿皮质兴奋性高,刺激能量过高需要注意TMS诱发癫痫可能); ②囟门关闭与否(未闭要特别注意避免机械损伤); ③外耳道生长(年龄小于2岁儿童需要特别注意保护听力)。儿童年龄小于2岁慎用[2]。

6 问题与展望

以往应用TMS的最大难题是如何精准刺激靶点。Karlström等[72]和Bohning等[74]于1998年将神经导航定位系统应用于TMS,通过与fMRI/CT等影像技术结合实现了可视化TMS,结合PET、SPECT、DTI、fMRI、磁共振波谱等技术可以更好地探讨TMS的治疗机制。目前存在主要问题包括: ①治疗效果的影响因素; ②各类疾病作用机制; ③rTMS与药物治疗的结合。

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